A MATHEMATICAL STRUCTURE OF THE TRANSFORMER

We follow the conventions and notation of Elhage et al. (2021) in describing our model. Here, we briefly recap their notation and examine it in our specific case.

We denote our hyperparameters as follows: $d_{vocab} = 113$ is the size of the input and output spaces (treating '=' separately), $d_{model} = 128$ is the width of the residual stream (i.e. embedding size), $d_{head} = 32$ is the size of query, key and value vectors for a single attention head, and $d_{mlp} = 512$ is the number of neurons.

We denote the parameters as follows: W_E (embedding layer); W_{pos} (positional embedding); W_Q^j (queries), W_K^j (keys), W_V^j (values), W_O^j (attention output) (the 4 weight matrices of head j in the attention layer); W_{in} and b_{in} for the input linear map of the MLP layer; W_{out} and b_{out} for the output linear map of the MLP layer; and W_U (unembedding layer). Note that we do not have biases in our embedding, attention layer or unembedding, and we do not tie the matrices for the embedding/unembedding layers.

We now describe the mathematical structure of our network. Note that loss is only calculated from the logits on the final token, and information only moves between tokens during the attention layer, so our variables from the end of the attention layer onwards only refer to the final token. We use t_i to denote the token in position i (as a one-hot encoded vector), p_i to denote the ith positional embedding, $x_i^{(0)}$ to denote the initial residual stream on token with index i, $A^{(i)}$ to denote the attention scores from $=$ to all previous tokens from head i, $x^{(1)}$ to denote the residual stream after the attention layer on the final token, MLP to denote the neuron activations in the MLP layer on the final token, $x^{(2)}$ the final residual stream on the final token, Logits the logits on the final token.

The logits are calculated via the following equations:

$$
\begin{aligned}
x_i^{(0)} &= W_E t_i + p_i \\
A^j &= \text{softmax}({x^{(0)}}^T {W_K^j}^T W_Q^j x_2^{(0)}) \\
x^{(1)} &= [\sum_j W_O^j W_V^j (x^{(0)} \cdot A^j)] + x_2^{(0)} \\
\text{MLP} &= \text{ReLU}(W_{in} x^{(1)}) \\
x^{(2)} &= W_{out} N + x^{(1)} = W_{out}\text{ReLU}(W_{in} x^{(1)}) + x^{(1)} \\
\text{Logits} &= W_U x^{(2)}
\end{aligned}
$$

As in Elhage et al. (2021), we refer to the term $W_O^j W_V^j (x^{(0)})$ as the OV circuit for head j.

A.1 EMPIRICAL MODEL SIMPLIFICATIONS

We make two empirical observations:

- The attention paid from '=' to itself is trivial. In practice, the average attention paid is 0.1% to 0.4% for each head, and ablating this does not affect model performance at all.
- The skip connection around the MLP layer is not important for the model's computation and can be ignored. Concretely, if we set it to zero or to its average (zero or mean ablation) then model accuracy is unchanged, and loss goes from $2.4 \cdot 10^{-7}$ to $9.12 \cdot 10^{-7}$ and $7.25 \cdot 10^{-7}$ respectively. This is a significant increase in loss, but from such a small baseline that we can still ignore it and reverse engineer the model's computation. (That being said, both the attention heads and the skip connection around them are crucial to the functioning of the model: zero ablating attention heads increases loss to 24.3, while zero ablating the skip connection around the attention heads increases loss to 19.1, both *significantly* worse than chance.)

A consequence of the first observation is that the attention is now a softmax over 2 elements, i.e. a sigmoid over the difference. And $x_2^{(0)}$ is constant, as it is independent of x and y, and the embedding

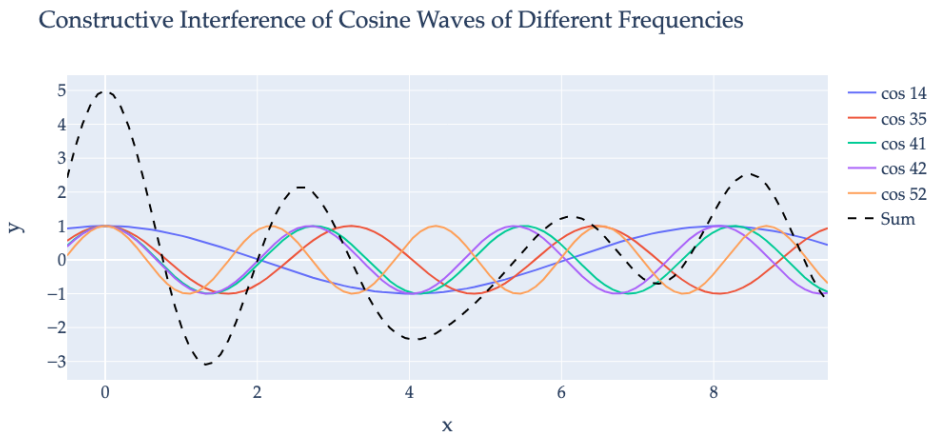

Figure 8: As discussed in Appendix B, while for every $k \in [0, ...P-1]$, $\cos\left(\frac{2k\pi}{P}x\right)$ achieves its maximum value (1) at $x = 0 \mod 113$, it still has additional peaks at different values that are close to the maximum value. However, by adding together cosine waves of the 5 keyfrequencies, the model constructs a periodic function where the value at $x = 0 \mod 113$ is significantly larger than its value anywhere else.

and positional embedding of '=' are fixed. So $A_0^j = \sigma\left({x_2^{(0)}}^T {W_Q^j}^T W_K(x_0^{(0)} - x_1^{(0)})\right)$ (and $A_1^j = 1 - A_0^j$)

A consequence of the second observation is that Logits $\approx W_U W_{out}$MLP, which we denote as $W_L = W_U W_{out}$. From the perspective of the network, W_L is the meaningful matrix, not either of its constituents, since they compose linearly.

B Why use constructive intereference?

As demonstrated in Section 4 and Appendix C.2.1, small transformers trained on this task use several different frequencies which they add together. The reason for this is to end up with a function whose value at $x = 0 \mod 113$ is significantly larger than any other x.

For example, consider the function $f_{14}(x) = \cos\left(\frac{2\pi \cdot 14}{113}x\right)$. This function has period 113 and is maximized at $x = 0 \mod 113$. However, other values of x cause this function to be close to 1: $f_{14}(8) = f_{14}(105) = 0.998$, $f_{14}(16) = f_{14}(89) = 0.994$, etc.

Now consider $f_{35}(x) = \cos\left(\frac{2\pi \cdot 35}{113}x\right)$. While this function also has period 113 and is maximized at $x = 0 \mod 113$, it turns out that $f_{35}(8) = f_{35}(105) = -0.990$. This means that by adding together f_{14} and f_{35}, we end up with a function that is not close to 1 at $x = 8 \mod 113$. Similarly, while $f_{35}(16) = 0.961$, $f_{52}(16) = -0.56$, and so adding a third frequency reduces the peak at $x = 16 \mod 113$.

We show the constructive interference resulting from the cosine waves for the five frequencies used by the mainline model in Figure 8.

C Supporting evidence for mechanistic analysis of modular arithmetic networks

C.1 Further analysis of the specific training run discussed in the paper

In this section, we provide additional evidence relating to the mainline model.

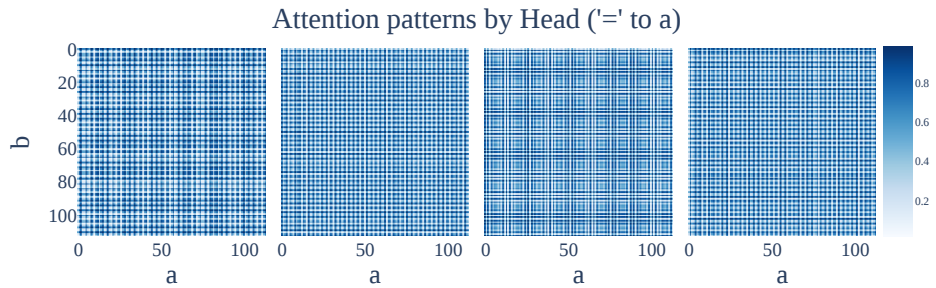

Figure 9: Attention patterns for each head, from the '=' token at the third sequence position to the a token at the first sequence position, as a heatmap over the inputs. All four attention heads exhibit striking periodicity.

Head	k	α^j	β^j	FVE
0	35	-0.26	-0.14	99.03%
1	42	0.27	-0.04	98.49%
2	52	0.29	-0.05	99.07%
3	42	-0.26	0.04	97.91%

Table 2: For each attention head, we show the pattern from '=' to a is well approximated by $0.5 + \alpha(\cos(w_k a) - \cos(w_k b)) + \beta(\sin(w_k a) - \sin(w_k b))$ and give the coefficients and fraction of variance explained for this approximation.

C.1.1 Periodicity in the activations of other attention heads

In Figure 9 we plot the attention patterns from the final token '=' to the first token a for all 4 attention heads, as a heatmap over the inputs a and b, as this is a scalar for each head. We observe a striking periodicity and further that heads 1 and 3 represent the same frequency while heads 0 and 2 are different.

As shown in Appendix A.1, the attention paid from '=' to itself is negligible, so $A_0^j = 1 - A_1^j$ and it suffices to plot attention to a.

C.1.2 Approximating attention heads with Sines and Cosines

Attention heads approximately compute degree-2 polynomials of a single frequency or are used to amplify W_E. In order to compute terms like $\cos(w_k(a+b))$, the model needs to compute the product of the sine and cosine embeddings output by W_E. As the attention heads are approximately bilinear (product of attention weights and OV circuit), they are a natural place to perform this computation. Indeed, for each head, the attention scores' Fourier transform is concentrated on a single frequency w_k. For two of the four heads, the corresponding OV circuit is concentrated on that same frequency. Moreover, the softmax mapping the attention scores to attention weights is in a regime where it behaves approximately linearly (and replacing it with a linear function actually improves performance). Thus the attention weights multiply with the OV output to create degree-2 polynomials of the frequency w_k, as would be needed for the cosine/sine addition formulas.

For the remaining two heads, their attention scores approximately sum to one and the OV circuits contain all five key frequencies, suggesting that they are used to increase the magnitude of key frequencies in the residual stream. We confirm all of these claims in Appendix C.1.3.

C.1.3 The attention pattern weights are well approximated by differences of sines and cosines of a single frequency.

The periodicity of the attention heads has a striking form—A_0^j is well approximated by $0.5 + \alpha^j(\cos(w_k a) - \cos(w_k b)) + \beta^j(\sin(w_k a) - \sin(w_k b))$, for some frequency w_k and constants α^j and β^j (which may differ for each head). Note further that this simplifies to $0.5 + \gamma(\cos(w_k(a + \theta)) - \cos(w_k(b + \theta)))$ for some constants γ and θ. We show the coefficients and fraction of variance explained in Table 1